Certified adaptive examination of peripheral blood smears: when information-gain stopping fails, and what can be guaranteed instead

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Abstract

Background and Objective: Automated smear examination increasingly decides which fields to inspect and when to stop. We asked what such a procedure can guarantee about unexamined material, and how many expert-verified cells are needed to certify whether the blast percentage exceeds a diagnostic threshold.

Methods: We reimplemented the information-saturation halting rule of InfoDPP-PAC, a recent patch-selection method, and compared it at matched cost with uniform sampling, a fixed order and confidence-based stopping on pre-registered composite specimens built from 28,280 expert-labelled cells with controlled rarity. We derived the minimax number of fields needed to certify the absence of rare evidence. For 10% and 20% blast thresholds we evaluated design-based, anytime-valid sequential tests with uniform and classifier-driven sampling, and tested their assumptions under acquisition shift on two external datasets, spatial clustering, unequal field sizes and reader disagreement in annotated bone-marrow regions. Analyses used exact binomial, McNemar and Wilcoxon tests and bootstrap intervals.

Results: The saturation rule's stopping point barely changed with rarity; with one critical field among 200 it missed 76% of positive specimens (uniform sampling of equal size: 85%). Without location information, any strategy must examine 95% of the fields to certify absence at a 5% error level. The blast classifier's sensitivity was 0.91 at 0.95 specificity in distribution and fell to 0.63 and 0.34 under acquisition shift, with specificity near 0.94. Valid tests kept false certification below 0.05 wherever their assumptions held, including under shift. Their cost was governed mainly by the margin to the threshold; near it, a 100-cell count was wrong in up to 37% of specimens, whereas valid procedures were mostly undetermined. Classifier-driven sampling needed 0.28–0.73 times the verified cells of uniform sampling in distribution, and 1.77 times under the strongest shift. Treating cells of sampled fields as independent draws exceeded the nominal error in 27 of 75 real specimens, and single-reader verification certified that reader's count, not the consensus.

Conclusions: Information saturation does not certify unexamined material. A classifier can reduce verification effort, but validity depends on the sampling design, the unit of analysis and its weighting, and the reliability of verification.

Keywords: adaptive sampling, sequential testing, anytime-valid inference, risk-limiting audit, blood smear, blast percentage

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1. Introduction

A peripheral blood smear contains far more cells than any examiner inspects; a manual differential counts 100 or 200 leukocytes, with well-known sampling variability [1]. Automated systems add a degree of freedom: they choose which fields to examine and when to stop. For decisions that hinge on rare cells, such as whether the blast percentage exceeds a classification threshold [2, 3], what matters is not classifier accuracy on the examined fields but what can be said about the fields that were not examined.

Stopping rules based on *information saturation* answer this question implicitly: they stop when further fields would add little information about the selector's representation of the specimen. InfoDPP-PAC [4], a recent preprint for whole-slide histopathology, selects patches greedily under a blended information-gain and relevance objective and stops when the largest remaining information gain falls below a concentration-corrected threshold. Its authors state that the certificate bounds residual information in the selection objective, not diagnostic performance. That distinction is our starting point.

Guarantees about partly examined finite populations come from a different literature. Risk-limiting audits test assertions about a population of ballots with a bounded probability of confirming a false assertion [5]; ALPHA [6] turns such assertions into bounded-mean tests for sampling without replacement, and Shekhar et al. [7] extended them to weighted, adaptive sampling with side information, anytime-valid through Ville's inequality [8, 9]. Whether these tools apply to smear examination, and whether their assumptions hold on real smear data, is examined here.

We address four research questions. RQ1: does information-saturation stopping fail systematically when decision-changing evidence is rare? RQ2: why does certifying the *absence* of rare evidence require near-exhaustive examination? RQ3: how many cells must be examined to certify a threshold decision under adaptive sampling? RQ4: where do the assumptions of anytime-valid audit methods break on real data? The study is methodological. Most specimens are composites of real single-cell images, and no result establishes clinical diagnostic validity.

2. Methods

2.1. Setting and estimands

A specimen has N fields; field i holds m_i cells, B_i of them blasts, $M = \sum_i m_i$, and $p = \sum_i B_i/M$. We distinguish *detection* (has a critical field, one with at least one blast, been examined?), *estimation* of p, and *threshold certification* (is $p > \theta$?). We use $\theta = 0.10$, the blast percentage at which the International Consensus Classification separates myelodysplastic syndrome with excess blasts from MDS/AML [3], and $\theta = 0.20$, the WHO 5th-edition threshold for acute myeloid leukaemia without defining genetic abnormalities [2]. Both classifications count blast equivalents, which our labels do not; the thresholds are analysed separately and are inputs to the procedure, not validated by it. An examiner selects fields sequentially, possibly using features x and past labels, and each selected field is *verified*: its cells' labels are revealed. All probabilities are over the examiner's randomisation for a fixed specimen, so every guarantee is finite-population and design-based.

2.2. Certifying the absence of rare evidence (RQ2)

Let C be the set of k critical fields, M_n the event that none of them is among n examined fields, and $h(N, k, n) = \binom{N-k}{n}/\binom{N}{n}$.

Proposition 1 (Uniform sampling)**.** *Under simple random sampling of n fields without replacement, $\mathbb{P}(M_n \mid C) = h(N, k, n)$. For $k = 1$ this equals $1 - n/N$, so $\mathbb{P}(M_n) \le \alpha$ if and only if $n \ge \lceil (1-\alpha)N \rceil$. For every $0 \le n \le N$, $[\max\{0, 1 - n/(N-k+1)\}]^k \le h \le (1 - n/N)^k$, so $n \ge N(1 - \alpha^{1/k})$ suffices.*

Assumption 1 (No location information)**.** Given the features x, every placement C with $|C| = k$ is possible.

Theorem 1 (Minimax lower bound)**.** *Under Assumption 1, every policy, adaptive or not, randomised or not, that examines at most n fields satisfies $\max_{|C|=k} \mathbb{P}_\pi(M_n \mid x, C) \ge h(N, k, n)$. Uniform sampling attains the bound, and every deterministic policy has worst-case miss probability 1 if $n \le N - k$.*

The proof (Supplementary Section S4) follows Yao's principle [10]. If k is unknown the worst case is $k = 1$, so certifying absence at level α needs $\lceil (1-\alpha)N \rceil$ fields. Informative features can let a search beat this bound on average, through a model of where evidence lies:

Proposition 2 (Ranked search)**.** *Let one field be critical with score $S_c \sim F_1$, and let the other $N-1$ scores be independent with distribution F_0. A scan of the n highest-scoring fields misses the critical field with probability $\mathbb{E}_{S_c}[\mathbb{P}(\mathrm{Bin}(N-1, 1-F_0(S_c)) \geq n)]$, which tends to $1 - \mathrm{TPR}(t_r)$ as $N \to \infty$ with $n/(N-1) \to r$, where t_r is the threshold with false-positive rate r.*

With $F_1 = F_0$ the formula returns $1 - n/N$. Certifying absence by ranking thus needs a sensitivity of about $1-\alpha$ at a false-positive rate of n/N, a property of the model that no sampling design supplies. The proposition is an average-case result under independence; it does not show that only ranking can help, and it is not a worst-case guarantee, because a ranked scan misses with certainty any critical field placed among the low-scoring fields. All statements were verified exactly (Supplementary Section S1.2).

2.3. Certifying a threshold assertion (RQ3)

Making "$p > \theta$" true requires more than θM blasts, which cannot hide in one field, so Theorem 1 does not apply. With equal field sizes and the assorter $a_j = b_j/(2\theta)$, where b_j is the blast fraction of the j-th sampled field, the specimen mean of a exceeds $1/2$ if and only if $p > \theta$, and the assertion becomes a one-sided bounded-mean test. Under uniform sampling of fields, ALPHA [6] gives a test supermartingale, and rejecting when it exceeds $1/\alpha$ bounds the probability of false certification by α at any data-dependent stopping time. For adaptive sampling, write $p = \sum_i w_i f_i$ with known weights w_i and $f_i \in [0,1]$. If field I_t is drawn from a predictable distribution q_t that is positive on every unsampled field, the observation $Z_t = w_{I_t} f_{I_t}/q_t(I_t)$ has conditional mean equal to the unsampled mass, which under $H_0\colon p \leq \theta$ is at most $\mu_t = \theta - \sum_{s<t} w_{I_s} f_{I_s}$. A betting process $\prod_s \{1 + \lambda_s(Z_s - \mu_s)\}$ with predictable, non-negative, bounded bets is then a test supermartingale whatever the classifier does [7]. We used the prop-MS sampler of that work, $q_t(i) \propto w_i S_i$ with a floor,

where S_i is the field's mean classifier score, and approximate-Kelly bets. Validity requires only known, positive selection probabilities and verified labels.

Averaging field proportions estimates $N^{-1}\sum_i B_i/m_i$, which is not p when field sizes differ.

Proposition 3 (Unequal field sizes)**.** *(a) If every m_i is known, $w_i = m_i/M$ and $f_i = B_i/m_i$ give $\sum_i w_i f_i = p$, so the weighted test is valid for $H_0 : p \le \theta$. (b) If only a bound $m_i \le m_{\max}$ is known, $a_i = (B_i - \theta m_i + \theta m_{\max})/m_{\max} \in [0,1]$ and $N^{-1}\sum_i a_i > \theta$ if and only if $p > \theta$, so equal-weight tests on a_i with null mean θ are valid.*

In (b) the margin shrinks by $\bar{m}/m_{\max}$, so a loose bound costs power.

Remark 1 (Imperfect verification). If the verifier's false blast calls number at most β_0 times the true non-blasts, its proportion satisfies $\tilde{p} \le p + \beta_0(1-p)$, so testing $\tilde{p} > \theta + \beta_0(1-\theta)$ is valid for $p > \theta$. Symmetrically, testing $\tilde{p} < \theta(1-\beta_1)$ is valid for $p < \theta$ if missed blasts number at most β_1 times the true blasts. The bound concerns the verifier, not the design.

We compared, on each specimen, ALPHA and betting with uniform sampling, betting with classifier-driven sampling (with and without a control variate), an invalid *uncorrected* design that ranks fields by score but analyses them as uniform draws, and conventional 100- and 200-cell counts.

2.4. Data

KU-Optofil [11] contains Sysmex DI-60 images of May–Grünwald–Giemsa-stained peripheral blood smears labelled by haematologists; we used 28,280 cells from 275 identified patients (13 classes) in five patient-grouped folds. Its 489 blasts come from only 11 patients. RV-PBS [12] provides 752 annotated cells (153 blasts) from phone-camera fields and was used only as an acquisition-shift probe. MLL23 [13] contains 41,621 peripheral-blood cells (Pappenheim stain, Metafer scanner), including 8,606 myeloblasts; we excluded atypical promyelocytes, which both classifications count as blast equivalents but our labels do not, smudge cells and normoblasts, leaving 36,529 cells without patient identifiers. BoneMarrowWSI [14, 15] provides pediatric bone-marrow aspirate slides with

228 exhaustively annotated regions of about 225 μm; we used the annotations, the label from each reading round, and the image pixels of the annotated regions.

2.5. Representation, teachers and specimens

Cells were embedded with a frozen DINOv2 ViT-S/14 encoder [16]. A logistic probe cross-fitted by patient fold reached an out-of-fold area under the ROC curve (AUC) of 0.979 and, at the threshold giving 95% specificity, a blast sensitivity of 0.906. The pre-registered *cell-level teacher* scores a field by its largest cell score; an exploratory *field-level teacher* is a logistic regression on the pooled field embedding that the selector sees. Part I composites have $N = 200$ fields of $m \in \{1, 5, 10, 20\}$ cells with mean-pooled embeddings and $k \in \{1, 2, 5, 10, 20\}$ critical fields of one blast each. Part III composites have 100 fields of 5 cells ($M = 500$) with p fixed exactly; exploratory extensions use M up to 4,000 and field sizes drawn from BoneMarrowWSI.

2.6. InfoDPP-PAC and baselines (RQ1)

No code has been released for InfoDPP-PAC. We implemented its objective, greedy blended gain and halting rule (Eqs. 3–7 of [4]) with the published weights, set the unreported constant $\rho = 1$, and calibrated the halting threshold on blast-free specimens of other folds so that the median stop was 16.3% of all fields. The original runs stopped at 16.3% of a 300-patch budget, about 2.4% of a slide according to the reported averages, so our setting gives InfoDPP-PAC roughly seven times its own examination share; a 2.4% run is also reported. A random seed of 5 fields fits the Gaussian process and is not counted as examined. Baselines were uniform sampling of the same size in the same specimen, a fixed field order, confidence-based stopping (stop at the first field with relevance ≥ 0.9 or after 30 consecutive fields ≤ 0.1), and exhaustive examination. Implementation details and all deviations are in Supplementary Sections S1.4 and S1.5.

2.7. Conditions for RQ4

The assumptions behind the guarantees were stressed in eight conditions (Table 3): composites in distribution, with strongly clustered blasts, with a background enriched in the classes most often mistaken for blasts, and with unequal field sizes drawn from marrow

regions; acquisition shift (RV-PBS, MLL23), with the classifier applied unchanged; real cell arrangements, with each BoneMarrowWSI patient as a specimen, 4×4 subregions of about $56\,\mu$m as fields and the specimen's own blast proportion as threshold ($\theta_s = p_s$, an exact null at which every certification is false); and a single reader as verifier, with the consensus as truth and field counts from that reader. Part I was also repeated on real image tiles of the annotated regions.

2.8. Pre-registration and statistical analysis

Hypotheses, decision rules, seeds and scripts were frozen in five git tags before the corresponding data were generated (Supplementary Section S5). "Confirmatory" means fresh Monte Carlo draws on a disjoint seed family over the same cell pool; only the external-shift and real-clustering hypotheses used data not analysed before. Later analyses used new seed families and are reported as exploratory. Miss rates were compared with exact McNemar tests on paired episodes, with Wilson intervals and t intervals on the five fold estimates. False-certification rates were tested against 0.05 with one-sided exact binomial tests and reported with Clopper–Pearson intervals. Costs were compared with paired Wilcoxon signed-rank tests with Holm correction and ratios of medians with bootstrap intervals. Real-data analyses bootstrap over patients. $\alpha = 0.05$ unless stated. The implementation was checked against the reference implementation of ALPHA and with 26 deterministic tests, including adversarial samplers and negative controls (Supplementary Section S1.4). AI assistance used for coding is declared at the end of the article; all code was verified with these checks.

3. Results

3.1. Information-saturation stopping (RQ1)

In 24,000 confirmatory episodes the median stop was 32–33 of 200 fields in every cell and varied by at most one field across rarity levels. This is an empirical finding for the tested settings: the halting statistic has no relevance term, but it depends on the embeddings and the selected set, which depend on the cells. The AUC with which the information-gain term separated critical from non-critical fields fell from 0.778 at $m = 1$

to 0.525 at $m = 20$, whereas the field-level teacher on the same pooled embeddings kept 0.978 to 0.827 (Fig. 1b). The evidence survived pooling; the halting signal did not use it.

With one critical field, InfoDPP-PAC missed it in 0.315, 0.681, 0.759 and 0.805 of specimens at $m = 1$, 5, 10 and 20, against 0.82–0.85 for uniform sampling of the same size, which matched $1 - n/N$ (Fig. 1a). The advantage shrank with dilution for every k, but the pre-registered claim of no advantage at $m \geq 10$ was not supported: the difference was -0.087 $[-0.107, -0.067]$ at $m = 10$, and -0.023 at $m = 20$ with a fold-level interval of $[-0.050, 0.003]$. Dilution, not rarity, governed the gap; rarity drove the absolute miss rate of every method. Two other saturation rules, pure information-gain sampling and k-centre coverage stopping, missed within 0.04 of InfoDPP-PAC in every cell. A scan ranked by the field-level teacher missed 0.05–0.37, within 0.02 of the values predicted by Proposition 2. At about 2.4% of the specimen, InfoDPP-PAC made 4–5 selections after its seed and missed 0.81–0.97 against 0.97–0.98 for uniform sampling, an advantage of -0.17 $[-0.27, -0.08]$ with single-cell fields and at most about 0.03 otherwise, whether or not the seed was counted as examined.

On real image tiles (3,040 tiles of about 56 μm from 99 patients; specimens of 16–32 tiles), InfoDPP-PAC stopped after 5.4 tiles on average. In the 24 specimens in which at most a quarter of the tiles were critical, it missed every critical tile in 0.80 of runs, against 0.70 for uniform sampling of the same size, a difference of $+0.105$ $[0.002, 0.203]$. It missed in at least 95% of runs in 15 of these specimens, as Theorem 1 predicts for a nearly deterministic selector.

3.2. Certifying absence (RQ2)

Figure 2 shows the fraction of fields that any strategy without location information must examine. With one critical field and $\alpha = 0.05$ it is 95% of the specimen for every N; with $k = 10$ critical fields it is still about a quarter.

3.3. Certified threshold decisions (RQ3)

In 12,000 confirmatory 500-cell specimens, the four valid designs kept false certification at $p = \theta$ between 0.001 and 0.022, whereas the uncorrected design certified falsely in 0.993

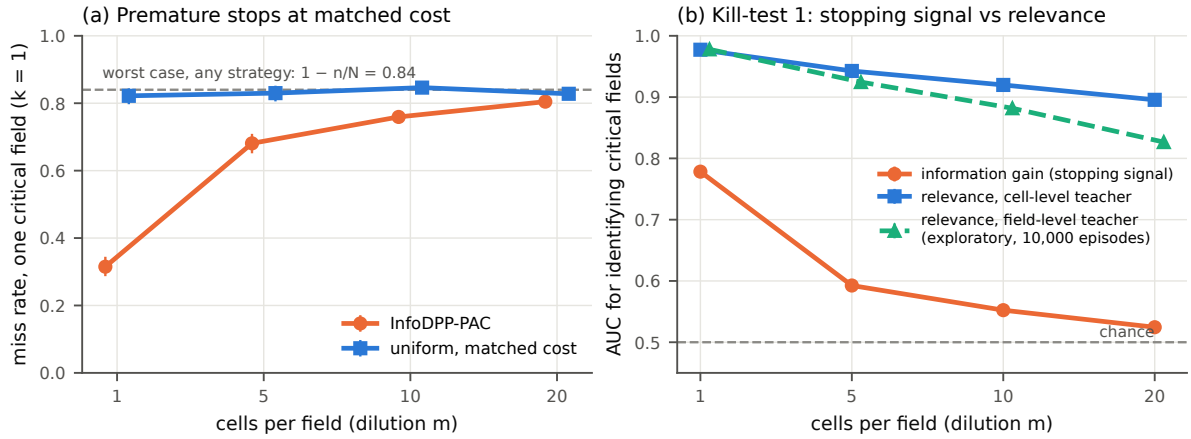


Figure 1: Information-saturation stopping (RQ1), confirmatory run. (a) Miss rate with one critical field for InfoDPP-PAC and for uniform sampling of the same size (95% Wilson intervals); dashed: the worst-case bound $1 - n/N$. (b) AUC of the information-gain term that drives stopping and of the cell-level teacher for ranking critical above non-critical fields; dashed: the field-level teacher on the same pooled embeddings as the selector (exploratory, 10,000 separate episodes).

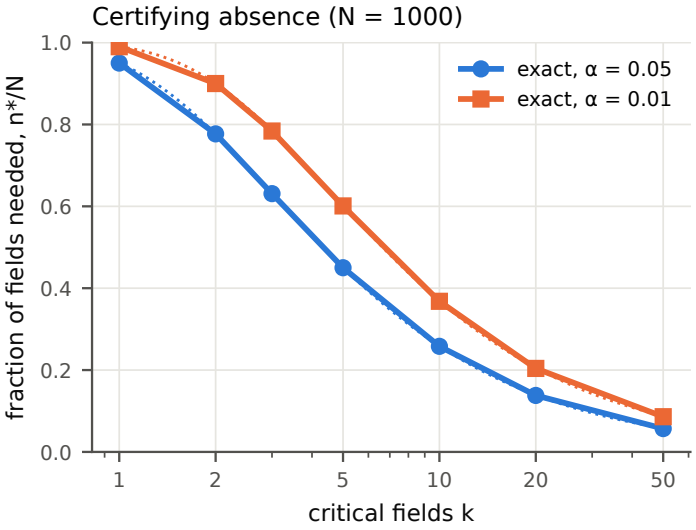


Figure 2: Certifying absence (RQ2). Fraction of fields n^*/N that any strategy without location information must examine to certify the absence of k critical fields ($N = 1000$). Solid: exact values; dotted: the sufficient closed form $1 - \alpha^{1/k}$.

($\theta = 0.10$) and 0.889 ($\theta = 0.20$) of boundary specimens (Fig. 3a). Classifier-driven betting needed fewer cells than uniform ALPHA at every $p > \theta$ (all Holm-adjusted $p < 10^{-59}$), for example 75 against 165 cells at $p = 0.15$, $\theta = 0.10$, and cost fell with the margin $|p - \theta|$ (Fig. 3b,c). Holding the test fixed, the classifier-driven sampler needed 0.28–0.73 times the cells of uniform sampling.

With specimens of up to 4,000 cells (exploratory; 17,600 specimens), validity was unchanged (false certification at most 0.040 at $\alpha = 0.05$ and 0.007 at $\alpha = 0.01$). Cost in cells was governed mainly by the margin (Table 1): at margins of 0.05 or more it changed by at most 40% between 1,000 and 4,000 cells and was within 20% of Wald's approximation $\log(1/\alpha)/\mathrm{KL}(p \,\|\, \theta)$ [17], whereas within 0.02 of the threshold it still grew with M, towards about 1,400 ($\theta = 0.10$) and 2,500 cells ($\theta = 0.20$). The near-census observed at $M = 500$ was therefore a small-specimen effect. All savings are counts of expert-verified cells; classifier-driven sampling still needs every field to be imaged and scored.

At fixed budgets (exploratory; 2,000-cell specimens), we compared three-way decisions: "above", "below" or "undetermined" (Table 2). Within 0.02 of the threshold a 100-cell count was wrong in 18–37% of specimens. Valid procedures were wrong in at most 0.7% of specimens off the threshold, but at those margins returned "undetermined" in 61–98% of specimens at budgets up to 500 cells.

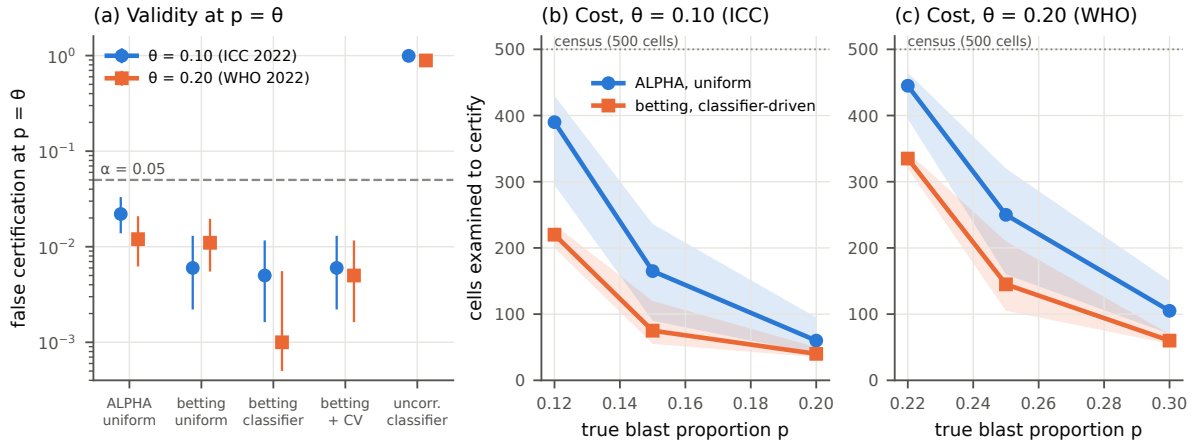


Figure 3: Certified threshold decisions (RQ3), confirmatory run (500-cell specimens, $\alpha = 0.05$, designs paired on the same specimen). (a) False certification at $p = \theta$ with 95% Clopper–Pearson intervals (log scale); CV: control variate; the uncorrected design analyses classifier-ranked fields as uniform draws. (b, c) Median cells to certification with interquartile bands; specimens that never certify count as 500 cells.

Table 1: Median cells to certify $p > \theta$ against specimen size M (exploratory; $\alpha = 0.05$; 400 specimens per cell; 5 cells per field). U: ALPHA with uniform sampling; C: betting with classifier-driven sampling; Approx.: $\log(1/\alpha)/\mathrm{KL}(p\|\theta)$. †Blast images reused, which affects only C.

		$M = 500$		$M = 1{,}000$		$M = 2{,}000$		$M = 4{,}000$		
θ	p	U	C	U	C	U	C	U	C	Approx.
0.10	0.12	390	220	655	420	1,040	660	1,583	798	1,425
0.10	0.15	165	80	205	80	220	85	245†	85†	245
0.10	0.20	65	40	68	40	65	40	75†	40†	67
0.20	0.22	435	335	810	660	1,338	1,300	2,033†	2,043†	2,455
0.20	0.25	250	145	303	175	338†	190†	420†	185†	406
0.20	0.30	100	60	115	65	120†	65†	125†	70†	106

Table 2: Three-way decisions after n individually sampled cells (exploratory; $M = 2{,}000$; 2,000 specimens per row; $\alpha = 0.05$). Conv.: share wrong for the conventional count. Exact: one-sided hypergeometric tests at $\alpha/2$, valid at this n only. Seq.: two one-sided sequential tests at $\alpha/2$, valid at every n. Entries for Exact and Seq. are shares correct / wrong; the rest is "undetermined".

		$n = 100$			$n = 500$	
θ	p	Conv.	Exact	Seq.	Exact	Seq.
0.10	0.08	0.18	0.08 / <0.01	0.04 / <0.01	0.39 / 0	0.15 / <0.01
0.10	0.12	0.33	0.08 / <0.01	0.02 / <0.01	0.34 / 0	0.15 / <0.01
0.10	0.15	0.10	0.30 / <0.01	0.12 / <0.01	0.96 / 0	0.86 / <0.01
0.20	0.18	0.24	0.07 / <0.01	0.02 / <0.01	0.23 / 0	0.09 / <0.01
0.20	0.22	0.37	0.05 / 0.01	0.02 / <0.01	0.25 / <0.01	0.08 / <0.01
0.20	0.25	0.14	0.18 / 0	0.06 / 0	0.87 / 0	0.59 / 0

3.4. Where the assumptions break (RQ4)

Table 3 summarises validity and cost across the eight conditions. Wherever their assumptions held, the design-based tests kept false certification at or below 0.031; every analysis without a design-based guarantee failed in at least one condition, often while looking acceptable in another.

Classifier error and acquisition shift. On KU-Optofil, blast sensitivity at 95% specificity varied across the 11 blast patients from 0.39 to 1.00. Applied unchanged, the classifier's sensitivity fell to 0.34 [0.25, 0.42] on RV-PBS and 0.63 [0.62, 0.64] on MLL23, while its false-positive rate barely rose (0.058 and 0.060, against 0.050). Every design stayed valid, but efficiency did not transfer: classifier-driven sampling needed 0.76 [0.70, 0.81] times the cells of uniform sampling on MLL23 at $p = 0.15$, and 1.77 times on RV-PBS at $p = 0.20$. A 50% uniform mixture stayed valid and reduced the RV-PBS excess to +23% [11, 30], missing the pre-registered bound of 20%.

Clustering and the unit of analysis. Under strong synthetic clustering, analysing the cells of sampled fields as independent draws certified falsely in up to 0.478 of boundary specimens, whereas field-level designs stayed valid. In 116 exhaustively annotated marrow regions of 75 patients, blasts were clustered but weakly (bias-corrected intraclass correlation 0.013 [0.002, 0.028] at the quadrant scale; Fig. 4b). On the real arrangements, the cell-level analysis gave 0.051 overall, close to α, yet exceeded 0.05 in 27 of 75 specimens (significantly in 21) and reached 0.335, whereas the weighted test of Proposition 3(a) stayed at or below 0.01 in every specimen. Validity is a statement about each specimen, and a pooled rate can hide failures.

Unequal field sizes. With the field sizes of marrow quadrants (12–29 cells, interquartile range), averaging field proportions certified falsely in 0.38 and 0.32 of boundary specimens even with blasts placed independently of size, and in every specimen with blasts enriched in small fields; both constructions of Proposition 3 stayed at or below 0.005.

Verifier error. The second reader called a blast in 1.1% [0.7, 1.7] of the cells the first reader called a counted non-blast, and a non-blast in 1.2% [0.8, 1.6] of the first reader's blasts; in a further 2.7% of cells only one reader counted the cell. With the first reader

as verifier (83 specimens), the reader's blast proportion differed from the consensus by a median of 0.005 (maximum 0.075), and the weighted test certified "$p > \theta_s$" falsely in 0.28 of runs and "$p < \theta_s$" in 0.63. The correction of Remark 1 with the readers' disagreement rates was not enough (0.16 and 0.42). A margin of 0.03 sufficed for "above" in this sample, but "below" still failed in three specimens at 0.05; these margins were selected on the same specimens, are direction-specific and are not a validated correction.

Table 3: Validity and cost across conditions (summary of stored results). Top: share of specimens in which "$p > \theta$" was certified at the exact null $p = \theta$ ($\alpha = 0.05$; larger of $\theta = 0.10$ and 0.20 where both were run; 400–1,000 specimens). BoneMarrowWSI: rate pooled over runs with $\theta_s = p_s$, with the largest specimen rate in brackets (75 and 83 specimens, 400 orders each). Bottom: median cells to certify $p = 0.15$ against $\theta = 0.10$; uncertified specimens count all their cells. Specimens hold 500 cells, about 1,300 with unequal fields, and a median of 105 real cells in BoneMarrowWSI. Stressed: the assumption the condition tests (unit: analysed unit equals sampled unit; target: cell-weighted p). C/X: confirmatory/exploratory run. —: not run or not applicable. [a]With unequal fields and in BoneMarrowWSI, ALPHA uses the bounded assorter and betting the known counts (Proposition 3). [b]Coincides with ALPHA for equal fields; 1.00 with blasts enriched in small fields. Single reader with a margin of 0.03: 0.0003 [0.013].

	Composites of KU-Optofil cells				External cells		BoneMarrowWSI	
	In distr.	Clustered	Confusable backgr.	Unequal fields	RV-PBS	MLL23	Real arrangem.	One reader
Stressed	—	*unit*	*classifier*	*target*	*classifier*	*transfer*	*geometry*	*verifier*
Status	C	C	X	X	C	X	X	X
False certification of "$p > \theta$" at $p = \theta$								
ALPHA, uniform[a]	0.022	0.025	0.031	0	—	—	0 [0]	—
Betting, uniform[a]	0.011	0.005	0.011	0.005	0.011	0.011	0.001 [0.010]	0.277 [1.00]
Betting, classifier-driven	0.005	0.003	0.007	0.001	0.001	0.001	—	—
Betting, 50% mixture	0.014	—	—	—	0.003	0.004	—	—
Ranked, analysed as uniform	0.993	—	—	—	—	—	—	—
Cells of fields as draws	0.041	0.478	—	—	—	—	0.051 [0.335]	—
Mean of field proportions[b]	—	—	—	0.379	—	—	0.062 [1.00]	—
100-cell count, "above"	0.427	0.388	—	—	—	—	—	—
Median cells to certify $p = 0.15$ against $\theta = 0.10$								
ALPHA, uniform[a]	165	345	160	1,031	—	—	—	—
Betting, uniform[a]	270	355	265	646	260	283	—	—
Betting, classifier-driven	75	55	105	297	265	215	—	—
Betting, 50% mixture	145	—	—	—	285	220	—	—

3.5. Pre-registered hypotheses

Of the 20 pre-registered hypotheses, 17 were supported (Table 4).

4. Discussion

4.1. What is guaranteed, and by what

Our results separate three quantities that are often reported together. *Model confidence* about examined fields says nothing about unexamined ones. *Information saturation*

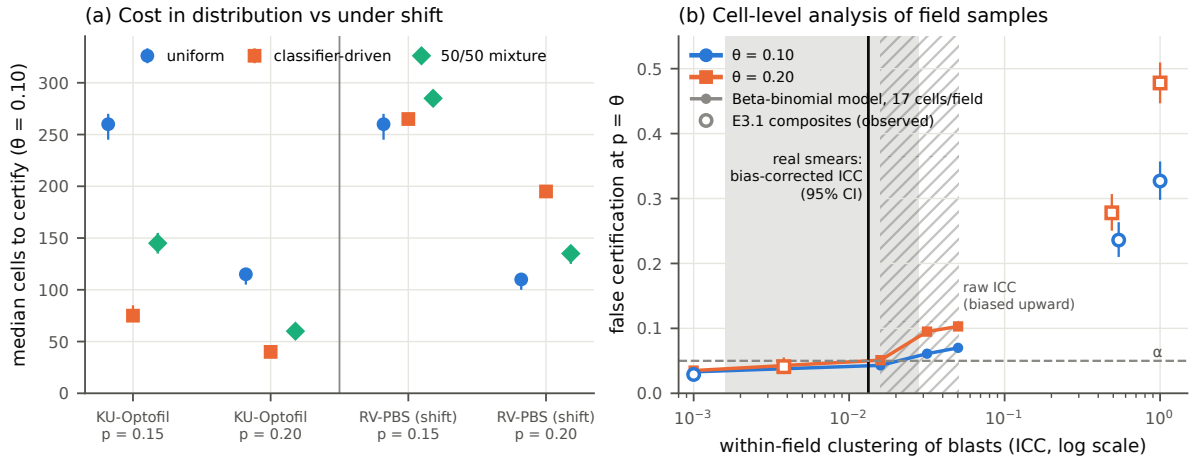


Figure 4: Where the assumptions break (RQ4). (a) Median cells to certify $p > 0.10$ in distribution (KU-Optofil) and under acquisition shift (RV-PBS); all designs keep false certification at or below 0.014. (b) False certification at $p = \theta$ when the cells of sampled fields are analysed as individual draws: composites with scattered or clustered arrangements (open markers) and a Beta-binomial model with 17 cells per field (lines). Solid band and line: bias-corrected clustering of real marrow regions (exploratory); hatched: the raw, upward-biased estimate, for comparison.

Table 4: Pre-registered hypotheses and verdicts. Type: E, an empirical claim; V, a check of the implementation or of a property that holds by theorem.

RQ	ID	Type	Hypothesis (abridged)	Verdict
1	H1a	E	InfoDPP-PAC stop point unchanged across rarity levels	supported
1	H1b	E	Advantage over matched-cost uniform sampling shrinks with dilution	supported (4/4)
1	H1c	E	No advantage (± 0.05) at $m \geq 10$, $k = 1$	**not supported**
2	H1d	V	Uniform miss rate equals $1 - n/N$	supported
1	KT1	E	Information gain and relevance are separable signals	supported
3	H3a	V	All four designs valid at both thresholds	supported
3	H3b	E	Uncorrected adaptive selection invalid	supported
3	H3c	E	Classifier-driven sampling cheaper than uniform	supported
3	H3d	E	Cost decreases with the margin	supported
4	H4a	E	Blast sensitivity drops under acquisition shift	supported
4	H4b	E	False-positive rate rises under shift	**not supported**
4	H4c	V	Certificate valid with shifted side information	supported
4	H5a	V	Field-level tests valid under clustering	supported
4	H5b	E	Cell-level analysis of field samples invalid under clustering	supported
4	H5c	E	Clustering raises the cost of valid certification	supported
4	H6a	V	Defensive mixture valid	supported
4	H6b	E	Mixture caps loss under shift below 20%	**not supported**
4	H6c	E	Mixture keeps $\geq 50\%$ of in-distribution saving	supported
4	H6d	E	Cost reversal under shift replicates	supported
4	H7a	E	Blasts spatially clustered in real smears	supported

concerns the selector's representation of the specimen; its association with decision-relevant evidence disappeared once cells were pooled into fields, although the pooled representation still carried the evidence. *Design-based certification* bounds the probability of a false threshold decision for the specific specimen, provided that selection probabilities are known and positive and that labels come from verification. Only the third is a guarantee about what was not examined, and it held in every condition we tested, specimen by specimen; the classifier affected only how quickly certification was reached. Table 5 lists what each guarantee assumes and what happened when the assumption was tested.

Table 5: What each guarantee assumes, and what happened when the assumption was tested.

Assumption	Needed for	Test and result
No location information	Theorem 1	Partly fails on composites; ranking helps on average, through a model (Proposition 2)
Selection probabilities known and positive	Validity	Uncorrected ranking: false certification 0.89–0.99
Analysis unit equals sampled unit	Validity	Cells as draws: up to 0.34 per real specimen, 0.33–0.48 under strong clustering
Target is the cell-weighted p	Validity	Averaging proportions: 0.32–0.38; Proposition 3 restores ≤ 0.005
Verified label equals truth	Validity for the true p	Single reader: 0.28 ("above"), 0.63 ("below")
Accurate classifier, exchangeable errors	Efficiency only	Validity held under shift (≤ 0.014); saving reversed (1.77 times the cells)

4.2. Relation to previous work

Rümke's analysis of the binomial variability of manual differentials [1] implies imprecision for low-frequency cell types; our conventional-count baseline shows what it means at a diagnostic threshold, where a 100-cell count forces an answer that is wrong in up to a third of specimens. For the absence of rare events, Proposition 1 and Theorem 1 are finite-population, design-based counterparts of the "rule of three" [18], extended to every strategy without location information. Classification accuracy, the focus of work on automated blast recognition [19], affected efficiency rather than validity in our results. The certificate of InfoDPP-PAC [4] bounds residual information in a selection objective and should not be read as a statement about unexamined diagnostic evidence. The anytime-valid tests of Stark [6] and Shekhar et al. [7], developed for election audits, transferred to

smear examination in our experiments; their assumptions were at risk in three places: the unit of analysis, the weighting of unequal fields and verifier error. Stratified tests [20] and prediction-powered inference [21, 22] were not implemented.

4.3. Practical implications

First, certify a proportion against a stated threshold rather than the absence of rare cells, which requires a near-census; the cost is governed mainly by the margin, and its dependence on specimen size fades once specimens are large and the margin is not small. Second, record selection probabilities, analyse the unit that was sampled, and weight fields by their cell counts. Third, allow and report "undetermined", the honest outcome of a few hundred cells near the threshold. Fourth, treat verifier error as part of the claim: relating a certificate to the true proportion requires a separately established bound on verifier error.

4.4. Limitations

Most specimens are composites from one laboratory, and all KU-Optofil blasts come from 11 patients, so cost and miss-rate comparisons are imprecise. The spatial, tile and verifier analyses use small, mostly high-blast pediatric bone-marrow specimens, not peripheral blood; they validate the method on real arrangements and reader labels but do not establish clinical validity, and the specimen-level analysis tests each specimen against its own blast proportion, not fixed diagnostic thresholds. The external datasets lack patient identifiers, so their intervals understate uncertainty; reading rounds are not identified experts; and several analyses were added after the pre-registered ones and are exploratory. InfoDPP-PAC's unreported hyperparameters were calibrated to a cost, although two other saturation rules gave the same result. We used a single frozen encoder, and costs count expert-verified cells only, not imaging or computing time.

5. Conclusions

Information saturation does not certify what remains unexamined, and without location information certifying the absence of rare evidence requires examining nearly the whole

specimen. Threshold decisions about the blast proportion can instead be certified with finite-sample, anytime-valid guarantees. A classifier can reduce the number of cells the expert must verify, but validity rests on the sampling design, the unit of analysis and weighting, and the reliability of verification. Fully annotated peripheral-blood smears and verification by identified haematologists are the most important next steps.

Ethics statement

This study used only publicly available, de-identified datasets released under open licences (KU-Optofil, MLL23 and BoneMarrowWSI under CC BY 4.0; RV-PBS as distributed by its authors). No new data were collected from human participants and no participant was contacted, so ethical approval and informed consent were not required for this secondary analysis. [To be confirmed against the University of Debrecen's requirements.]

CRediT authorship contribution statement

Qudrat Ullah: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Balázs Harangi:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

All code, the pre-registration with its git history, the audit trail and the analysis outputs are available at https://github.com/Afridi-7/certified-adaptive-pbs; a release will be archived on Zenodo with a DOI. KU-Optofil, BoneMarrowWSI-PediatricLeukemia and MLL23 are public datasets under CC BY 4.0. RV-PBS is publicly distributed by its authors without a stated licence and was used for research only.

Supplementary material

The supplementary file gives the full methods (Section S1, including the InfoDPP-PAC reimplementation and all deviations), the complete results with all sensitivity and exploratory analyses (Sections S2 and S3), the proof of Theorem 1 (Section S4) and the pre-registration audit trail (Section S5).

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[To be added.]

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

The research questions and study design come from the first author's research proposal. During the preparation of this work the authors used Claude (Anthropic) as an assistant for coding, checking calculations and references, and editing the text. The authors reviewed and edited all output and take full responsibility for the content of the published article.

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